

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

May 2026

SP9 - Enterprise Risk Management

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) Atra Tatra Sarvatra (ATS) is a multi-national conglomerate with multiple subsidiary companies operating in many sectors including, but not limited to, banking, life insurance, general insurance, and actuarial consulting within the group.

You lead the in-house risk and actuarial consulting practice responsible for advising group companies / subsidiaries in all actuarial and risk-related matters. As such, at various times, you are called upon to offer consultation to each of these different subsidiaries.

The Board of ATS is evaluating whether the group's ERM framework effectively integrates risk appetite, capital allocation, and strategic decision-making across subsidiaries.

- i) You are first called upon to advise their Central Risk Function (CRF) which has been recently set up. The CRF is in the process of coming up with their Key Risk Indicators (KRIs). Explain what KRIs are, and list some of their desirable traits. Also suggest at least two suitable KRIs (each corresponding to a different risk type) each for its banking and insurance subsidiaries for two different types of risks. [6]
- ii) The CRF has also been charged with studying Emerging Risks. Explain what emerging risks are, and describe their characteristics. Also cite suitable examples of such risks relevant to the group companies from the recent past or otherwise, along with their impact and potential mitigation. [10]
- iii) Another mandate to the CRF is to manage alignment of interests between stakeholders and identify the related risks. Provide four examples of risks arising from misalignment of interests between different stakeholders. [4]
- iv) Your next assignment is to advise the general insurance subsidiary. The remuneration sub-committee of the Board of this general insurance company has proposed the following uniform compensation structure for its entire executive leadership (all CXOs):
 - A base salary in cash (of value X per annum, paid monthly)
 - A quarterly cash bonus (of value X/4 per quarter) payable if and only if the company meets its quarterly revenue target
 - An annual stock option award (worth X/2 per annum, paid annually), which can be exercised two years from the date of award

Describe the advantages of and potential challenges associated with such a compensation structure, including its evaluation from a risk culture / alignment perspective. Also include any changes you would like to suggest to the structure. [9]

- v) Define any six of the following accounting ratios and explain their usefulness for assessing the risks faced by a general insurance company:
 - Asset utilisation ratio
 - Return on capital employed
 - Gross profit margin
 - Current ratio
 - Asset cover
 - Income gearing
 - Dividend yield
 - Payout ratio

[9]

vi) A new colleague in the Risk and Capital Modelling team has joined the general insurance company. Her prior experience is in a similar role at a bank. Explain to her the Solvency II framework highlighting the key similarities with the Basel accords. Also briefly comment on the differences in the two frameworks. [6]

vii) Your next stop is at the Risk division of the life insurance company. For the life insurance company, state (with an appropriate example for each) the various ways of transferring risks that the company is likely to be exposed to. Also explain the key considerations while making such risk transfer decisions, with a suitable example each. [12]

viii) Interest rate risk is a key risk for a life insurer. Describe some techniques that the life insurer can use to manage its direct and indirect exposures to this risk. Discuss briefly how these techniques interact with capital requirements. [8]

ix) Your next engagement is with the investment managers of the bond portfolio for each of the group subsidiaries. Each bond portfolio includes a wide variety of bonds (government as well as corporate issuers, varied maturities, denominated in different currencies, zero-coupon as well as coupon-paying, fixed as well as floating coupons, etc). Each manager currently has the cashflow model outlining the cashflows emanating from the portfolio (if the bonds were to be held till their respective maturity), and computing the net present value using an appropriate risk discount rate. Describe how and why each of the following might be performed:

- Sensitivity analysis
- Scenario analysis
- Stress testing

Include up to three examples for each of these activities. Also explain how outputs from these analyses would be used in ORSA and strategic decision-making. [15]

x) The final stop in the journey pertains to capital and its allocation across entities. VaR and TVaR are key metrics in terms of capital measurement and modelling. Explain VaR and TVaR and compare their suitability for capital measurement and modelling. [6]

xi) You have been asked to present why the capital required under a company's regulatory regime might differ from the capital required based on the company's own internal model. Discuss some of the key reasons for the differences and the implications thereof. [5]

xii) Given below is the standalone capital for each of the three regions where ATS operates:

	Region P	Region Q	Region R
Revenue	40	30	50
Undiversified capital	62	40	88
Diversification benefit	20	10	32
Capital requirement	42	30	56

The group-level capital requirement is 96. In addition, some simulations of capital required for pairwise combinations of regions are provided below:

Total requirement for	
P + Q only	56
Q + R only	70
P + R only	82

Allocate capital to each using the pro rata approach 1 (proportional to standalone capital requirement), the pro rata approach 2 (proportional to revenue) and the marginal approach. Also comment on the advantages and limitations of each of these approaches. Additionally, comment (with reasons) on which method is the most appropriate for performance measurement.

[All figures are in ATS's headquarters' local currency in millions.]

[10]

[100 Marks]
